Section B

Question				Marking details	AO1	AO2	AO3	Total	Maths	Prac
					Marks available					
7.	(a)	(i)		van der Waals forces are stronger in iodine than in chlorine / more van der Waals forces in iodine than in chlorine (1) iodine has more electrons than chlorine (1) therefore more energy needed to overcome forces / higher melting temperature (1)	3			3		
		(ii)		aluminium has more valence electrons than sodium therefore stronger metallic bonds	1			1		
		(iii)		silicon has a giant molecular structure, phosphorus only has weak forces between the molecules	1			1		
	(b)			nitrogen is higher since it only has unpaired 2p electrons, O has two unpaired and two paired 2p electrons / N $1s^2\ 2s^2\ 2p^3$, O $1s^2\ 2s^2\ 2p^4$ (1) repulsion between paired electrons makes it easier to remove one of the electrons / takes more energy to remove unpaired electron (1)	2			2		
	(c)			percentage abundance 121 = 57%, 123 = 43% (1) accept 121 = 28.5, 123 = 21.5 if divided by 50 in calculation relative atomic mass = 121.9 (1)		2		2	2	

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(d)			^{90}Sr and ^{228}Th not suitable because half-life is too long so they remain in the body for a long time (1) ^{210}At and ^{228}Th not suitable because α-radiation is most ionising /causes most damage in the body (1) ^{99}Tc suitable because short half-life and γ-radiation does not stay in body (1)			3	3		
				Question 7 total	**7**	**2**	**3**	**12**	**2**	**0**